Guna Sai Kiran Nekkanti

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Modem Systems Engineer @ Qualcomm
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M.Tech in Electronics and Communication Engineering
Github | Website
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PROFESSIONAL SUMMARY

Modem Systems Engineer at Qualcomm and M.Tech Gold Medalist from IISc Bangalore, specializing in wireless communications, 5G/6G systems, and RIS-assisted wireless networks.

Experience

❒ **Modem Systems Engineer** *Jul. 2026 – Present*
Qualcomm Website
- ∗ Contributing to modem system design and analysis for next-generation cellular technologies.
- ∗ Developing system-level solutions and performance evaluation methodologies.
- ∗ Collaborating on wireless communication algorithms and modem features.

❒ **Teaching Assistant** *2025 – 2026*
Indian Institute of Science (IISc) Website
- ∗ Teaching Assistant for Random Processes and Digital Communication courses at IISc.
- ∗ Assisted with tutorials, assignments, and evaluations.

❒ **FPGA Design Engineer** *Dec. 2023 – Jun. 2024*
Internship | Bitmapper Integration Technologies Website
- ∗ Integrated CMOS image sensors with Xilinx FPGA platforms for image and video processing.
- ∗ Performed IP verification and FPGA pipeline design using Vivado, Vitis, and HLS.
- ∗ Worked with Spartan-7, Artix-7, and Virtex-7 boards and developed memory modules using C/C++.

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
MTech(ECE)	Indian Institute of Science Banglore	9.3	2024-2026
BTech(ECE)	Indian Institute of Information Technology, Nagpur	8.25	2020-2024
Intermediate	Tirumala Junior College	9.75	2018-2020

Technical Skills

- **Programming**: MATLAB, Python, C/C++, Verilog
- **Wireless Communications**: 5G NR, RIS, OFDM, Channel Estimation, Resource Scheduling
- **Tools**: MATLAB, Simulink, CST Studio Suite, ADS, Vivado, Vitis, Git, Linux, LaTeX
- **Protocols**: AXI, UART, SPI, I2C

Projects

❒ **RIS-Assisted 5G NR Systems: Theory and Experiments** *Aug. 2024 – Jun. 2026*
M.Tech Thesis | IISc Bangalore GitHub | IEEE ICC | IEEE TCOM
- ∗ Developed and experimentally validated RIS-assisted 5G NR wireless systems.
- ∗ Proposed opportunistic scheduling with randomized RIS configurations.
- ∗ Evaluated performance through simulations and real-time experiments.

❒ **Broadband FSS-Based Absorber** *Jul. 2023 – May 2024*
B.Tech Thesis | IIIT Nagpur GitHub | Electromagnetics (Taylor & Francis)
- ∗ Designed a broadband polarization-independent FSS absorber using 3D printing.
- ∗ Achieved 8.5–29.2 GHz absorption through CST, ADS, and MATLAB simulations.
- ∗ Explored applications in electromagnetic shielding and stealth technologies.

Achievements

- **Institute Gold Medalist**, M.Tech (ECE), IISc Bangalore.
- Secured **AIR 982** in GATE 2024 (EC).
- Qualified GATE 2023 (EC).

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